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JEE MAIN · CHEMISTRY · CHAPTER-WISE PYQS

Practical Organic Chemistry

JEE Main Chemistry · Previous Year Questions with Answer Key

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QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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MCQ (Single Correct Answer) 88 questions

Q1. JEE Main 2026 (Online) 6th April Evening Shift

Open solution

Given below are two statements :

Statement I : A mixture of $C_{12}H_{22}O_{11}$ (sugar) and NaCl can be separated by dissolving sugar in alcohol, due to differential solubility.

Statement II : Rose essence from rose petals is seperated by steam distillation due to its high volatility and insolubility in H_2O.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q2. JEE Main 2026 (Online) 6th April Morning Shift

Open solution

R_f value for 2-methylpropene in a solvent system (Ethyl acetate + ether) is 0.42 . 2-methylpropene is treated with dilute H_2SO_4 to give major organic product (X). R_f value for (X) in the same solvent system under identical condition will be:

(A) 0.42 (B) 0.82 (C) 0.62 (D) 0.12

Q3. JEE Main 2026 (Online) 5th April Evening Shift

Open solution

Match List - I with List - II.

List - I **Purification technique**		**List - II** **Used to separate**	
A.	Simple distillation	I.	Steam volatile compound
B.	Fractional distillation	II.	Two liquids with large difference in boiling points
C.	Steam distillation	III.	Liquid decomposing at its boiling point
D.	Distillation under reduced pressure	IV.	Two liquids with close boiling points

Choose the correct answer from the options given below :

(A) A-II, B-III, C-I, D-IV (B) A-II, B-IV, C-I, D-III

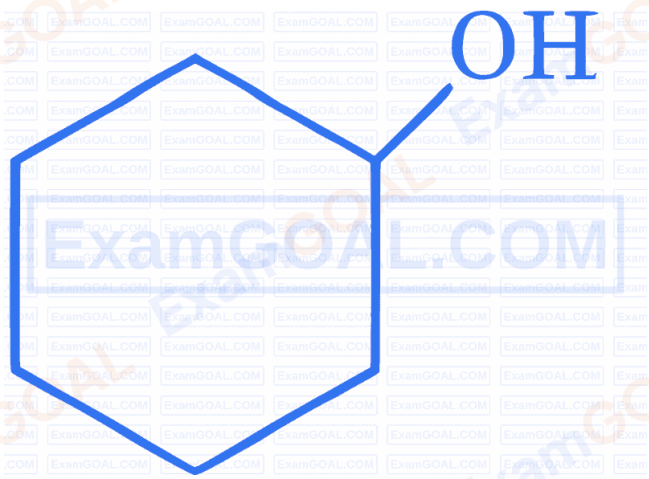
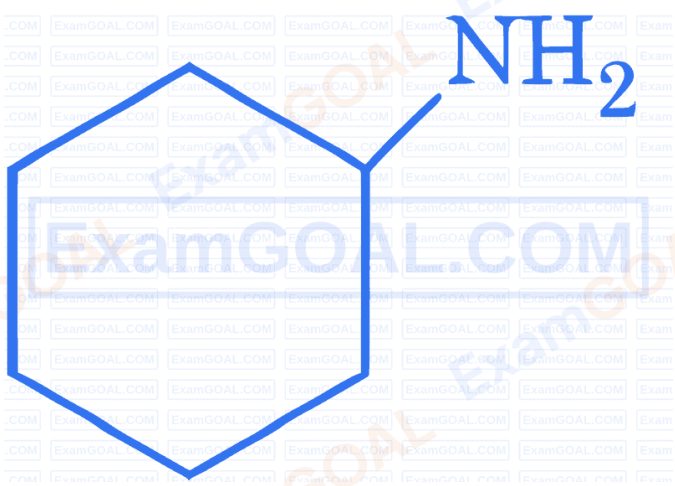
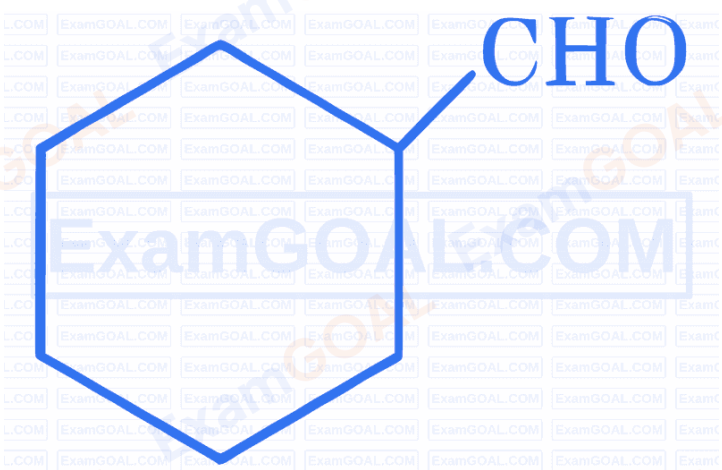
(C) A-II, B-IV, C-III, D-I (D) A-IV, B-III, C-II, D-I

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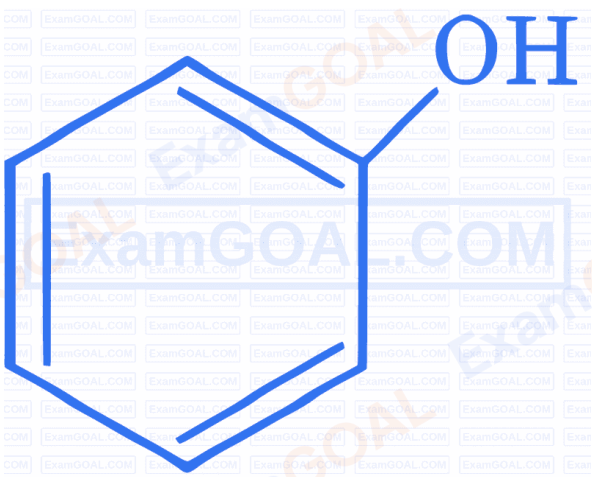
Q4. JEE Main 2026 (Online) 4th April Evening Shift

Open solution

Match the LIST-I with LIST-II

List - I		List - II	
	Compound		**Test**
A.		I.	Hinsberg's reagent test
B.		II.	Phthalein dye test
C.		III.	Lucas test

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D.		IV.	Tollen's test

Choose the correct answer from the options given below :

(A) A-III, B-I, C-IV, D-II

(B) A-III, B-IV, C-I, D-II

(C) A-I, B-III, C-II, D-IV

(D) A-I, B-II, C-III, D-IV

Q5. JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

Complete combustion of X g of an organic compound gave 0.25 g of CO_2 and 0.12 g of H_2O. If the % of carbon is 25% and of hydrogen is 4.89%, then $X =$ ___ $\times 10^{-3}$ g. (Nearest integer)

(Molar mass of C, H and O are 12, 1 and 16 g mol^{-1} respectively.)

(A) 273 (B) 27 (C) 2730 (D) 227

Q6. JEE Main 2026 (Online) 2nd April Morning Shift

Open solution

Given below are two statements :

Statement (I) : 1,2,3-Trihydroxypropane can be separated from water by simple distillation.

Statement (II) : An azeotropic mixture cannot be separated by fractional distillation.

In the light of the above statements, choose the **correct answer** from the options given below :

(A) Both **Statement I** and **Statement II** are true

(B) Both **Statement I** and **Statement II** are false

(C) **Statement I** is true but **Statement II** is false

(D) **Statement I** is false but **Statement II** is true

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Q7. JEE Main 2026 (Online) 2nd April Morning Shift

Open solution

Match **List - I** with **List - II**.

	List - I **Mixture of Compounds**		**List - II** **Reagent used to distinguish**
A	Diethyl amine + Ethyl amine	I	Bromine water
B	Acetaldehyde + Acetone	II	$CHCl_3$ + KOH, Δ
C	Ethanol + Phenol	III	Neutral $FeCl_3$
D	Benzoic acid + Cinnamic acid	IV	Ammoniacal silver nitrate

Choose the **correct** answer from the options given below :

(A) A-IV, B-II, C-I, D-III

(B) A-IV, B-II, C-III, D-I

(C) A-II, B-IV, C-I, D-III

(D) A-II, B-IV, C-III, D-I

Q8. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

A student has been given 0.314 g of an organic compound and asked to estimate Sulphur. During the experiment, the student has obtained 0.4813 g of barium sulphate. The percentage of sulphur present in the compound is _____. (Given Molar mass in g mol^{-1} S: 32, $BaSO_4$: 233)

(A) 63.15%

(B) 21.05%

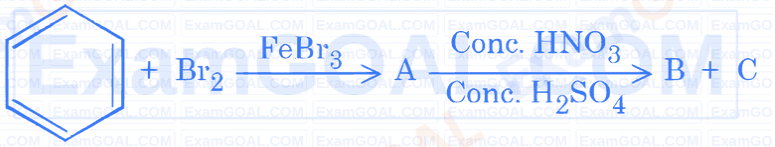
(C) 48.24%

(D) 42.10%

Q9. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

Method used for separation of mixture of products (B and C) obtained in the following reaction is



(A) simple distillation

(B) sublimation

(C) fractional distillation

(D) steam distillation

Q10. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution

In Carius method 0.2425 g of an organic compound gave 0.5253 g silver chloride. The percentage of chlorine in the organic compound is

(A) 87.65%

(B) 53.58%

(C) 37.57%

(D) 34.79%

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Q11. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

Match List - I with List - II.

List - I Functional group (detection)		List - II Change observed during detection	
A.	Unsaturation (Baeyer's test)	I.	Red colour appears
B.	Alcoholic group (Ceric ammonium nitrate test)	II.	Silver mirror appears
C.	Aldehyde group (Tollen's reagent)	III.	Violet colour appears
D.	Phenolic group ($FeCl_3$ test)	IV.	Discharge of pink colour

(A) A-IV, B-I, C-II, D-III
(B) A-III, B-IV, C-I, D-II
(C) A-III, B-IV, C-II, D-I
(D) A-IV, B-III, C-II, D-I

Q12. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

Identify the correct statements :

A. Hydrated salts can be used as primary standard.

B. Primary standard should not undergo any reaction with air.

C. Reactions of primary standard with another substance should be instantaneous and stoichiometric.

D. Primary standard should not be soluble in water.

E. Primary standard should have low relative molar mass.

Choose the correct answer from the options given below :

(A) D and E Only
(B) A, B and E Only
(C) A, B, C and E Only
(D) A, B and C Only

Q13. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

When 1 g of compound (X) is subjected to Kjeldahl's method for estimation of nitrogen, 15 mL 1 M H_2SO_4 was neutralized by ammonia evolved. The percentage of nitrogen in compound (X) is :

(A) 0.21 **(B)** 21 **(C)** 42 **(D)** 0.42

Q14. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

By usual analysis, 1.00 g of compound (X) gave 1.79 g of magnesium pyrophosphate. The percentage of phosphorus in compound (X) is : (nearest integer)

(Given, molar mass in g mol^{-1}: O = 16, Mg = 24, P = 31)

(A) 40 **(B)** 30 **(C)** 20 **(D)** 50

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Q15. JEE Main 2026 (Online) 21st January Morning Shift

In Carius method, 0.75 g of an organic compound gave 1.2 g of barium sulphate, find percentage of sulphur (molar mass $32\ \mathrm{g\ mol^{-1}}$). Molar mass of barium sulphate is $233\ \mathrm{g\ mol^{-1}}$.

(A) 4.55% (B) 16.48% (C) 10.30% (D) 21.97%

Open solution

Q16. JEE Main 2025 (Online) 7th April Evening Shift

A mixture of 1 g each of chlorobenzene, aniline, and benzoic acid is dissolved in 50 mL ethyl acetate and placed in a separating funnel. 5 M NaOH (30 mL) was added in the same funnel. The funnel was shaken vigorously and then kept aside. The ethyl acetate layer in the funnel contains :

(A) chlorobenzene and aniline
(B) benzoic acid and aniline
(C) benzoic acid
(D) benzoic acid and chlorobenzene

Open solution

Q17. JEE Main 2025 (Online) 4th April Evening Shift

Match List - I with List - II

	List - I (Separation of)		**List - II (Separation Technique)**
(A)	Aniline from aniline-water mixture	(I)	Simple distillation
(B)	Glycerol from spent-lye in soap industry	(II)	Fractional distillation
(C)	Different fractions of crude oil in petroleum industry	(III)	Distillation at reduced pressure
(D)	Chloroform-Aniline mixture	(IV)	Steam distillation

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
(B) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
(C) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)
(D) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

Open solution

Q18. JEE Main 2025 (Online) 4th April Evening Shift

A toxic compound " A " when reacted with NaCN in aqueous acidic medium yields an edible cooking component and food preservative " B ". " B " is converted to " C " by diborane and can be used as an additive to petrol to reduce emission. "C" upon reaction with oleum at $140^{\circ}\mathrm{C}$ yields an inhalable anesthetic " D ". Identify " A ", " B ", " C " & " D ", respectively :

(A) Methanol; acetic acid; ethanol; diethyl ether
(B) Ethanol; acetonitrile; ethylamine; ethylene
(C) Methanol; formaldehyde; methyl chloride; chloroform
(D) Acetaldehyde; 2-hydroxypropanoic acid; propanoic acid; dipropyl ether

Open solution

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Q19. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

In Dumas' method for estimation of nitrogen 0.4 g of an organic compound gave 60 mL of nitrogen collected at 300 K temperature and 715 mm Hg pressure. The percentage composition of nitrogen in the compound is (Given : Aqueous tension at $300\ \mathrm{K} = 15\ \mathrm{mmHg}$)

(A) 15.71% (B) 17.46% (C) 7.85% (D) 20.95%

Q20. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

In Dumas' method for estimation of nitrogen, 0.5 gram of an organic compound gave 60 mL of nitrogen collected at 300 K temperature and 715 mm Hg pressure. The percentage composition of nitrogen in the compound (Aqueous tension at $300\ \mathrm{K} = 15\ \mathrm{mmHg}$) is_____%.

(A) 20.87 (B) 12.57 (C) 1.257 (D) 18.67

Q21. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Match List - I with List - II.

List - I (Purification technique)		**List - II (Mixture of organic compounds)**	
(A)	Distillation (simple)	(I)	Diesel + Petrol
(B)	Fractional distillation	(II)	Aniline + Water
(C)	Distillation under reduced pressure	(III)	Chloroform + Aniline
(D)	Steam distillation	(IV)	Glycerol + Spent-lye

Choose the correct answer from the options given below :

(A) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
(B) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
(C) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)
(D) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

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Q22. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

Given below are two statements :

Statement (I): In partition chromatography, stationary phase is thin film of liquid present in the inert support.

Statement (II): In paper chromatography, the material of paper acts as a stationary phase.

In the light of the above statements, choose the **correct** answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

Q23. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

Given below are two statements:

Statement I: In Lassaigne's test, the covalent organic molecules are transformed into ionic compounds.

Statement II: The sodium fusion extract of an organic compound having N and S gives prussian blue colour with $FeSO_4$ and $Na_4[Fe(CN)_6]$

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

Q24. JEE Main 2024 (Online) 9th April Evening Shift

Open solution

Match List I with List II

	LIST I (Test)		**LIST II (Observation)**
A.	Br_2 water test	I.	Yellow orange or orange red precipitate formed
B.	Ceric ammonium nitrate test	II.	Reddish orange colour disappears
C.	Ferric chloride test	III.	Red colour appears
D.	2, 4 - DNP test	IV.	Blue, Green, Violet or Red colour appears

Choose the correct answer from the options given below:

(A) A-II, B-III, C-IV, D-I

(B) A-III, B-IV, C-I, D-II

(C) A-I, B-II, C-III, D-IV

(D) A-IV, B-I, C-II, D-III

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Q25. JEE Main 2024 (Online) 9th April Morning Shift

Open solution

Identify the incorrect statements regarding primary standard of titrimetric analysis.

(A) It should be purely available in dry form.

(B) It should not undergo chemical change in air.

(C) It should be hygroscopic and should react with another chemical instantaneously and stoichiometrically.

(D) It should be readily soluble in water.

(E) $KMnO_4$ & $NaOH$ can be used as primary standard.

Choose the correct answer from the options given below :

(A) (C) and (E) only **(B)** (A) and (B) only **(C)** (B) and (E) only **(D)** (C) and (D) only

Q26. JEE Main 2024 (Online) 8th April Evening Shift

Open solution

Match List I with List II

	LIST I (Test)		**LIST II (Identification)**
A.	Baeyer's test	I.	Phenol
B.	Ceric ammonium nitrate test	II.	Aldehyde
C.	Phthalein dye test	III.	Alcoholic-OH group
D.	Schiff's test	IV.	Unsaturation

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(I), (C)-(II), (D)-(III) **(B)** (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(C) (A)-(II), (B)-(III), (C)-(IV), (D)-(I) **(D)** (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

Q27. JEE Main 2024 (Online) 6th April Morning Shift

Open solution

Which of the following statements are correct?

A. Glycerol is purified by vacuum distillation because it decomposes at its normal boiling point.

B. Aniline can be purified by steam distillation as aniline is miscible in water.

C. Ethanol can be separated from ethanol water mixture by azeotropic distillation because it forms azeotrope.

D. An organic compound is pure, if mixed M.P. remains the same.

Choose the most appropriate answer from the options given below :

(A) A, C, D only **(B)** B, C, D only **(C)** A, B, D only **(D)** A, B, C only

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Q28. JEE Main 2024 (Online) 1st February Evening Shift

Lassaigne's test is used for detection of :

(A) Phosphorous and halogens only

(B) Nitrogen, Sulphur and Phosphorous only

(C) Nitrogen, Sulphur, phosphorous and halogens

(D) Nitrogen and Sulphur only

Open solution

Q29. JEE Main 2024 (Online) 1st February Morning Shift

In Kjeldahl's method for estimation of nitrogen, $CuSO_4$ acts as :

(A) catalytic agent (B) hydrolysis agent (C) reducing agent (D) oxidising agent

Open solution

Q30. JEE Main 2024 (Online) 1st February Morning Shift

Given below are two statements :

Statement (I) : Potassium hydrogen phthalate is a primary standard for standardisation of sodium hydroxide solution.

Statement (II) : In this titration phenolphthalein can be used as indicator.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(A) Both Statement I and Statement II are correct

(B) Statement I is incorrect but Statement II is correct

(C) Both Statement I and Statement II are incorrect

(D) Statement I is correct but Statement II is incorrect

Open solution

Q31. JEE Main 2024 (Online) 31st January Evening Shift

The fragrance of flowers is due to the presence of some steam volatile organic compounds called essential oils. These are generally insoluble in water at room temperature but are miscible with water vapour in vapour phase. A suitable method for the extraction of these oils from the flowers is -

(A) distillation (B) steam distillation

(C) distillation under reduced pressure (D) crystallisation

Open solution

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Q32. JEE Main 2024 (Online) 31st January Morning Shift

Open solution

Match List I with List II

	List - I (Technique)		List - II (Application)
(A)	Distillation	(I)	Separation of glycerol from spent-lye
(B)	Fractional distillation	(II)	Aniline - Water mixture
(C)	Steam distillation	(III)	Separation of crude oil fractions
(D)	Distillation under reduced pressure	(IV)	Chloroform - Aniline

Choose the correct answer from the options given below:

(A) A-I, B-II, C-IV, D-III **(B)** A-II, B-III, C-I, D-IV

(C) A-IV, B-I, C-II, D-III **(D)** A-IV, B-III, C-II, D-I

Q33. JEE Main 2024 (Online) 30th January Morning Shift

Open solution

The Lassiagne's extract is boiled with dil HNO_3 before testing for halogens because,

(A) Silver halides are soluble in HNO_3.

(B) $AgCN$ is soluble in HNO_3.

(C) Na_2S and $NaCN$ are decomposed by HNO_3.

(D) Ag_2S is soluble in HNO_3.

Q34. JEE Main 2024 (Online) 29th January Evening Shift

Open solution

Chromatographic technique/s based on the principle of differential adsorption is / are

A. Column chromatography

B. Thin layer chromatography

C. Paper chromatography

Choose the most appropriate answer from the options given below:

(A) B only **(B)** A only **(C)** A & B only **(D)** C only

Q35. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

In chromyl chloride test for confirmation of Cl^- ion, a yellow solution is obtained. Acidification of the solution and addition of amyl alcohol and 10% H_2O_2 turns organic layer blue indicating formation of chromium pentoxide. The oxidation state of chromium in that is

(A) +6 **(B)** +5 **(C)** +3 **(D)** +10

Q36. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

Appearance of blood red colour, on treatment of the sodium fusion extract of an organic compound with $FeSO_4$ in presence of concentrated H_2SO_4 indicates the presence of element/s

(A) Br **(B)** S **(C)** N and S **(D)** N

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Q37. JEE Main 2024 (Online) 27th January Evening Shift

The technique used for purification of steam volatile water immiscible substances is :

(A) Fractional distillation

(B) Distillation

(C) Fractional distillation under reduced pressure

(D) Steam distillation

Open solution

Q38. JEE Main 2023 (Online) 10th April Evening Shift

In Carius tube, an organic compound 'X' is treated with sodium peroxide to form a mineral acid 'Y'.

The solution of $BaCl_2$ is added to 'Y' to form a precipitate 'Z'. 'Z' is used for the quantitative estimation of an extra element. 'X' could be

(A) Cytosine (B) Methionine (C) Chloroxylenol (D) A nucleotide

Open solution

Q39. JEE Main 2023 (Online) 10th April Morning Shift

Given below are two statements:

Statement I : Aqueous solution of $K_2Cr_2O_7$ is preferred as a primary standard in volumetric analysis over $Na_2Cr_2O_7$ aqueous solution.

Statement II : $K_2Cr_2O_7$ has a higher solubility in water than $Na_2Cr_2O_7$.

In the light of the above statements, choose the correct answer from the options given below:

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are true

Open solution

Q40. JEE Main 2023 (Online) 8th April Evening Shift

A compound 'X' when treated with phthalic anhydride in presence of concentrated H_2SO_4 yields 'Y'. 'Y' is used as an acid/base indicator. 'X' and 'Y' are respectively

(A) Toluidine, Phenolphthalein (B) Carbolic acid, Phenolphthalein

(C) Salicylaldehyde, Phenolphthalein (D) Anisole, methyl orange

Open solution

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Q41. JEE Main 2023 (Online) 8th April Evening Shift

Open solution

Given below are two statements:

Statement I : Methyl orange is a weak acid.

Statement II : The benzenoid form of methyl orange is more intense/deeply coloured than the quinonoid form.

In the light of the above statement, choose the most appropriate answer from the options given below:

(A) Both Statement I and Statement II are incorrect

(B) Statement I is incorrect but Statement II is correct

(C) Both statement I and Statement II are correct

(D) Statement I is correct but Statement II is incorrect

Q42. JEE Main 2023 (Online) 6th April Morning Shift

Open solution

Match List I with List II

	LIST I Element detected		**LIST II Reagent used / Product formed**
A.	Nitrogen	I.	$Na_2[Fe(CN)_5NO]$
B.	Sulphur	II.	$AgNO_3$
C.	Phosphorous	III.	$Fe_4[Fe(CN)_6]_3$
D.	Halogen	IV.	$(NH_4)_2MoO_4$

Choose the correct answer from the options given below:

(A) A-III, B-I, C-IV, D-II

(B) A-II, B-I, C-IV, D-III

(C) A-IV, B-II, C-I, D-III

(D) A-II, B-IV, C-I, D-III

Q43. JEE Main 2023 (Online) 1st February Evening Shift

Open solution

Given below are two statements :

Statement I : Sulphanilic acid gives esterification test for carboxyl group.

Statement II : Sulphanilic acid gives red colour in Lassaigne's test for extra element detection.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are correct

(B) Statement I is correct but Statement II is incorrect

(C) Both Statement I and Statement II are incorrect

(D) Statement I is incorrect but Statement II is correct

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Q44. JEE Main 2023 (Online) 1st February Evening Shift

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : An aqueous solution of KOH when used for volumetric analysis, its concentration should be checked before the use.

Reason (R) : On aging, KOH solution absorbs atmospheric CO_2.

In the light of the above statements, choose the correct answer from the options given below :

(A) (A) is correct but (R) is not correct

(B) (A) is not correct but (R) is correct

(C) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(D) Both (A) and (R) are correct and (R) is the correct explanation of (A)

Q45. JEE Main 2023 (Online) 31st January Evening Shift

Open solution

In Dumas method for the estimation of N_2, the sample is heated with copper oxide and the gas evolved is passed over :

(A) Copper oxide **(B)** Copper gauze **(C)** Pd **(D)** Ni

Q46. JEE Main 2023 (Online) 29th January Evening Shift

Open solution

An indicator 'X' is used for studying the effect of variation in concentration of iodide on the rate of reaction of iodide ion with H_2O_2 at room temp. The indicator 'X' forms blue coloured complex with compound 'A' present in the solution. The indicator 'X' and compound 'A' respectively are :

(A) Methyl orange and H_2O_2 **(B)** Methyl orange and iodine

(C) Starch and H_2O_2 **(D)** Starch and iodine

Q47. JEE Main 2023 (Online) 29th January Morning Shift

Open solution

Compound that will give positive Lassaigne's test for both nitrogen and halogen is :

(A) $NH_2OH.HCl$ **(B)** $N_2H_4.HCl$ **(C)** $CH_3NH_2.HCl$ **(D)** NH_4Cl

Q48. JEE Main 2022 (Online) 27th July Morning Shift

Open solution

In Carius method of estimation of halogen, $0.45\ g$ of an organic compound gave $0.36\ g$ of $AgBr$. Find out the percentage of bromine in the compound.

(Molar masses : $AgBr = 188\ g\ mol^{-1}; Br = 80\ g\ mol^{-1}$)

(A) 34.04% **(B)** 40.04% **(C)** 36.03% **(D)** 38.04%

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Q49. JEE Main 2022 (Online) 25th July Evening Shift

In base vs. acid titration, at the end point methyl orange is present as

(A) quinonoid form
(B) heterocyclic form
(C) phenolic form
(D) benzenoid form

Open solution

Q50. JEE Main 2022 (Online) 30th June Morning Shift

The reagent neutral ferric chloride is used to detect the presence of _____

(A) sulphide ion and alcoholic –OH group.
(B) acetate ion and phenolic –OH group.
(C) sulphide ion and phenolic –OH group.
(D) acetate ion and alcoholic –OH group.

Open solution

Q51. JEE Main 2022 (Online) 28th June Morning Shift

The formula of the purple colour formed in Lassaigne's test for sulphur using sodium nitroprusside is :

(A) $NaFe[Fe(CN)_6]$
(B) $Na[Cr(NH_3)_2(NCS)_4]$
(C) $Na_2[Fe(CN)_5(NO)]$
(D) $Na_4[Fe(CN)_5(NOS)]$

Open solution

Q52. JEE Main 2022 (Online) 26th June Morning Shift

Given below are two statements :

Statement I : In 'Lassaigne's Test', when both nitrogen and sulphur are present in an organic compound, sodium thiocyanate is formed.

Statement II : If both nitrogen and sulphur are present in an organic compound, then the excess of sodium used in sodium fusion will decompose the sodium thiocyanate formed to give NaCN and Na_2S.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are correct.
(B) Both Statement I and Statement II are incorrect.
(C) Statement I is correct but Statement II is incorrect.
(D) Statement I is incorrect but Statement II is correct.

Open solution

Q53. JEE Main 2022 (Online) 25th June Evening Shift

During halogen test, sodium fusion extract is boiled with concentrated HNO_3 to

(A) remove unreacted sodium
(B) decompose cyanide or sulphide of sodium
(C) extract halogen from organic compound
(D) maintain the pH of extract.

Open solution

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Q54. JEE Main 2021 (Online) 27th August Evening Shift

Which one of the following tests used for the identification of functional groups in organic compounds does not use copper reagent?

(A) Barfoed's test
(B) Seliwanoff's test
(C) Benedict's test
(D) Biuret test for peptide bond

Open solution

Q55. JEE Main 2021 (Online) 27th August Morning Shift

Acidic ferric chloride solution on treatment with excess of potassium ferrocyanide gives a Prussian blue coloured colloidal species. It is :

(A) $Fe_4[Fe(CN)_6]_3$
(B) $K_5Fe[Fe(CN)_6]_2$
(C) $HFe[Fe(CN)_6]$
(D) $KFe[Fe(CN)_6]$

Open solution

Q56. JEE Main 2021 (Online) 26th August Morning Shift

Which one of the following complexes is violet in colour?

(A) $[Fe(CN)_6]^{4-}$
(B) $[Fe(SCN)_6]^{4-}$
(C) $Fe_4[Fe(CN_6)]_3 \cdot H_2O$
(D) $[Fe(CN)_5NOS]^{4-}$

Open solution

Q57. JEE Main 2021 (Online) 20th July Evening Shift

In Carius method, halogen containing organic compound is heated with fuming nitric acid in the presence of :

(A) HNO_3
(B) $AgNO_3$
(C) $CuSO_4$
(D) $BaSO_4$

Open solution

Q58. JEE Main 2021 (Online) 20th July Morning Shift

An inorganic Compound 'X' on treatment with concentrated H_2SO_4 produces brown fumes and gives dark brown ring with $FeSO_4$ in presence of concentrated H_2SO_4. Also Compound 'X' gives precipitate 'Y', when its solution in dilute HCl is treated with H_2S gas. The precipitate 'Y' on treatment with concentrated HNO_3 followed by excess of NH_4OH further gives deep blue coloured solution, Compound 'X' is :

(A) $Co(NO_3)_2$
(B) $Pb(NO_2)_2$
(C) $Cu(NO_3)_2$
(D) $Pb(NO_3)_2$

Open solution

Q59. JEE Main 2021 (Online) 18th March Morning Shift

Reagent, 1-napthylamine and sulphanilic acid in acetic acid is used for the detection of

(A) N_2O
(B) NO_3^-
(C) NO
(D) NO_2^-

Open solution

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Q60. JEE Main 2021 (Online) 16th March Evening Shift

Open solution

Match List - I with List - II :

	List - I Test/Reagents/Observation(s)		List - II Species detected
(a)	Lassaigne's Test	(i)	Carbon
(b)	Cu(II) oxide	(ii)	Sulphur
(c)	Silver nitrate	(iii)	N, S, P, and halogen
(d)	The sodium fusion extract gives black precipitate with acetic acid and lead acetate	(iv)	Halogen Specifically

The correct match is :

(A) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

(B) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)

(C) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(D) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

Q61. JEE Main 2021 (Online) 26th February Evening Shift

Open solution

Seliwanoff test and Xanthoproteic test are used for the identification of _____ and _____ respectively.

(A) ketoses, proteins

(B) proteins, ketoses

(C) aldoses, ketoses

(D) ketoses, aldoses

Q62. JEE Main 2021 (Online) 26th February Morning Shift

Open solution

Which of the following is 'a' FALSE statement ?

(A) Carius method is used for the estimation of nitrogen in an organic compound.

(B) Kjeldahl's method is used for the estimation of nitrogen in an organic compound.

(C) Carius tube is used in the estimation of sulphur in an organic compound.

(D) Phosphoric acid produced on oxidation of phosphorus present in an organic compound is precipitated as $Mg_2P_2O_7$ by adding magnesia mixture.

Q63. JEE Main 2021 (Online) 25th February Evening Shift

Open solution

Which of the following compound is added to the sodium extract before addition of silver nitrate for testing of halogens?

(A) Nitric acid

(B) Sodium hydroxide

(C) Ammonia

(D) Hydrochloric acid

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Q64. JEE Main 2020 (Online) 2nd September Evening Slot

If you spill a chemical toilet cleaning liquid on your hand, your first aid would be

(A) aqueous NaOH (B) aqueous $NaHCO_3$ (C) aqueous NH_3 (D) vinegar

Open solution

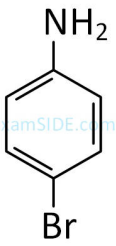
Q65. JEE Main 2020 (Online) 2nd September Morning Slot

In Carius method of estimation of halogen, 0.172 g of an organic compound showed presence of 0.08 g of bromine. Which of these is the correct structure of the compound?

(A) $H_3C - CH_2 - Br$

(B) $H_3C - Br$

(C)



(D)



Open solution

Q66. JEE Main 2020 (Online) 9th January Evening Slot

A, B and C are three biomolecules. The results of the tests performed on them are given below:

	Molisch's Test	Barfoed Test	Biuret Test
A	Positive	Negative	Negative
B	Positive	Positive	Negative
C	Negative	Negative	Positive

A, B and C are respectively :

(A) A = Lactose, B = Glucose, C = Albumin (B) A = Lactose, B = Fructose, C = Alanine

(C) A = Lactose, B = Glucose, C = Alanine (D) A = Glucose, B = Fructose, C = Albumin

Open solution

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Q67. JEE Main 2019 (Online) 12th April Morning Slot

An organic compound 'A' is oxidized with Na_2O_2 followed by boiling with HNO_3. The resultant solution is then treated with ammonium molybdate to yield a yellow precipitate.
Based on above observation, the element present in the given compound is:

Open solution

(A) Fluorine (B) Phosphorus (C) Nitrogen (D) Sulphur

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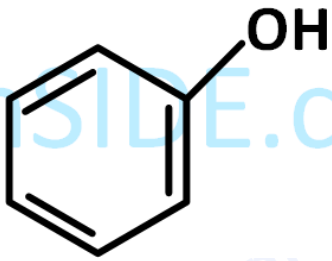
Q68. JEE Main 2019 (Online) 9th April Morning Slot

Open solution

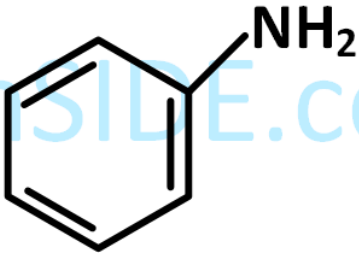
The organic compound that gives following qualitative analysis is :

	Test	Inference
(a)	Dil. HCl	Insoluble
(b)	NaOH solution	soluble
(c)	Br_2/water	Decolourization

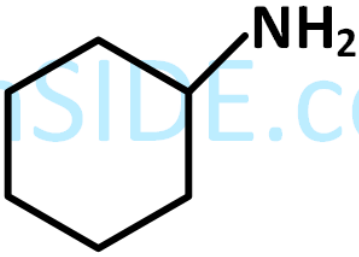
(A)



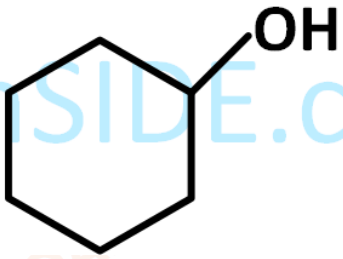
(B)



(C)



(D)



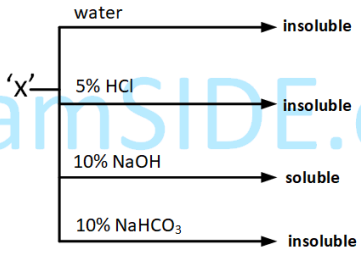
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Q69. JEE Main 2019 (Online) 8th April Morning Slot

Open solution

An organic compound 'X' showing the following solubility profile is :



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(A) o-Toluidine (B) Benzamide (C) Oleic acid (D) m-Cresol

Q70. JEE Main 2019 (Online) 8th April Morning Slot

Open solution

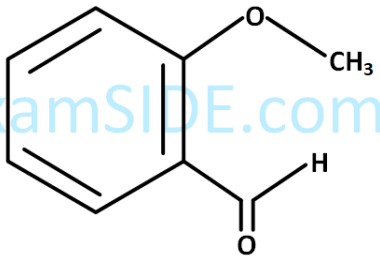
An organic compound neither reacts with neutral ferric chloride solution nor with Fehling solution. It, however, reacts with Grignard reagent and gives positive iodoform test. The compound is :

(A)



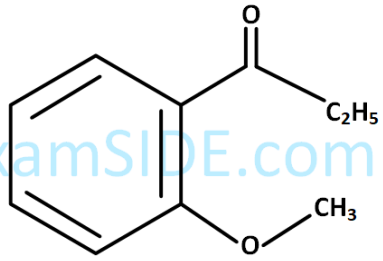
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(B)



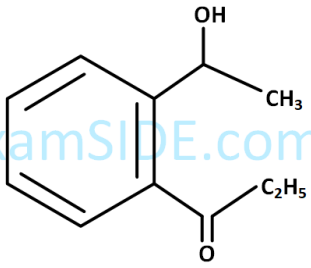
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(C)



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(D)



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Q71. JEE Main 2019 (Online) 10th January Evening Slot

Open solution

The correct match between item 'I' and item 'II' is :

Item 'I' (compound)		Item 'II' (reagent)	
(A)	Lysine	(P)	1-naphthol
(B)	Furfural	(Q)	ninhydrin
(C)	Benzylalcohol	(R)	$KMnO_4$
(D)	Styrene	(S)	Ceric ammonium nitrate

(A) (A) $\rightarrow$ (Q); (B) $\rightarrow$ (R); (C) $\rightarrow$ (S); (D) $\rightarrow$ (P)

(B) (A) $\rightarrow$ (Q); (B) $\rightarrow$ (P); (C) $\rightarrow$ (S); (D) $\rightarrow$ (R)

(C) (A) $\rightarrow$ (R); (B) $\rightarrow$ (P); (C) $\rightarrow$ (Q); (D) $\rightarrow$ (S)

(D) (A) $\rightarrow$ (Q); (B) $\rightarrow$ (P); (C) $\rightarrow$ (R); (D) $\rightarrow$ (S)

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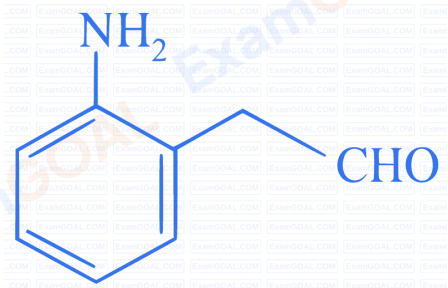
Q72. JEE Main 2019 (Online) 9th January Evening Slot

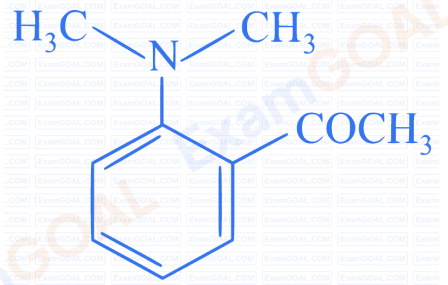
Open solution

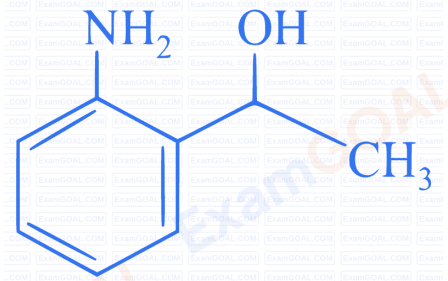
The tests performed on compound X and their inferences are :

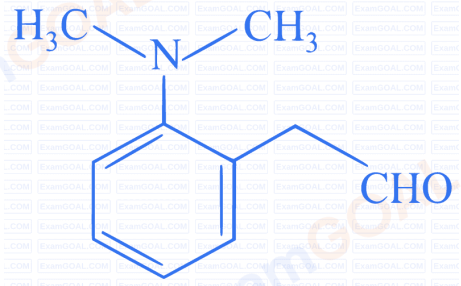
	Test	Inference
(a)	2, 4 - DNP test	Coloured precipitate
(b)	Idoform test	Yellow precipitate
(c)	Azo-dye test	No dye formation

Compound 'X' is :

(A) 

(B) 

(C) 

(D) 

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Q73. JEE Main 2019 (Online) 9th January Morning Slot

The correct match between Item-I and Item-II is :

Open solution

Item-I (drug)		Item-II (test)	
A	Chloroxylenol	P	Carbylamine test
B	Norethindrone	Q	Sodium hydrogen carbonate test
C	Sulphapyridine	R	Ferric chloride test
D	Penicillin	S	Bayer's test

(A) A $\rightarrow$ R; B $\rightarrow$ P; C $\rightarrow$ S; D $\rightarrow$ Q

(B) A $\rightarrow$ Q; B $\rightarrow$ S; C $\rightarrow$ P; D $\rightarrow$ R

(C) A $\rightarrow$ R; B $\rightarrow$ S; C $\rightarrow$ P; D $\rightarrow$ Q

(D) A $\rightarrow$ Q; B $\rightarrow$ P; C $\rightarrow$ S; D $\rightarrow$ R

Q74. JEE Main 2018 (Online) 16th April Morning Slot

For standardizing NaOH solution, which of the following is used as a primary standard ?

Open solution

(A) Ferrous Ammonium Sulfate

(B) dil. HCl

(C) Oxalic acid

(D) Sodium tetraborate

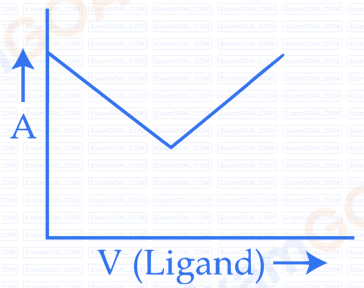
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Q75. JEE Main 2018 (Online) 16th April Morning Slot

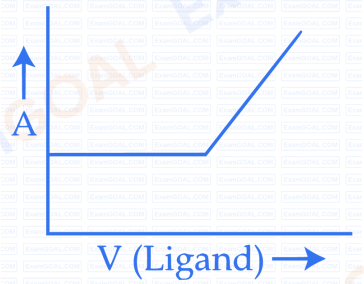
Open solution

In a complexometric titration of metal ion with ligand
M(Metal ion) + L(Ligand) $\rightarrow$ C(Complex) end point is estimated spectrophoto - metrically (through light absorption). If 'M' and 'C' do not absorb light and only 'L' absorbs, then the titration plot between absorbed light (A) versus volume of ligand 'L' (V) would look like :

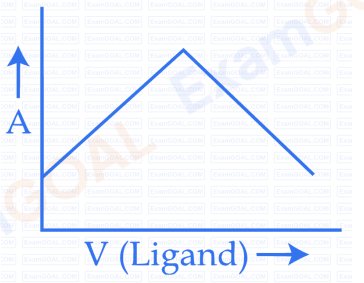
(A)



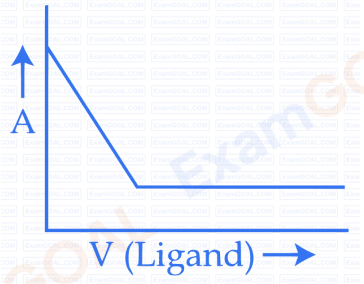
(B)



(C)



(D)



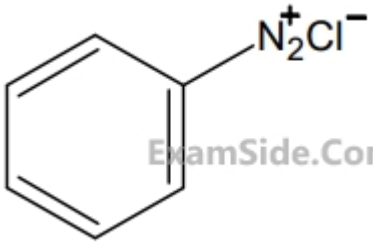
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Q76. JEE Main 2018 (Offline)

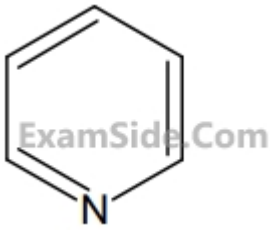
Open solution

Which of the following compounds will be suitable for Kjeldahl's method for nitrogen estimation?

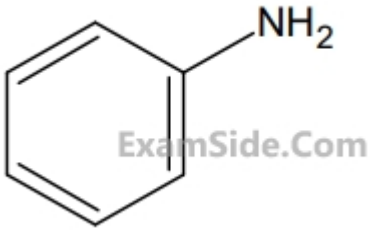
(A)



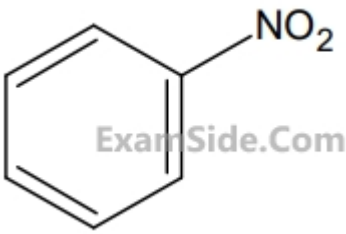
(B)



(C)



(D)



Q77. JEE Main 2017 (Offline)

Open solution

Sodium salt of an organic acid 'X' produces effervescence with conc. H_2SO_4. 'X' reacts with the acidified aqueous $CaCl_2$ solution to give a white precipitate which decolourises acidic solution of $KMnO_4$. 'X' is :

(A) CH_3COONa (B) $Na_2C_2O_4$ (C) C_6H_5COONa (D) HCOONa

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Q78. JEE Main 2016 (Online) 10th April Morning Slot

Open solution

Sodium extract is heated with concentrated HNO_3 before testing for halogens because :

(A) Silver halides are totally insoluble in nitric acid.

(B) Ag_2S and AgCN are soluble in acidic medium.

(C) S_2^- and CN^-, if present, are decomposed by conc. HNO_3 and hence do not interfere in the test.

(D) Ag reacts faster with halides in acidic medium.

Q79. JEE Main 2015 (Online) 11th April Morning Slot

Open solution

Match the organic compounds in **column - I** with the Lassaigne's test results in **column - II** appropriately :

Column - I		**Column - II**	
(A)	Aniline	(i)	Red color with $FeCl_3$
(B)	Benzene sulfonic acid	(ii)	Violet color with sodium nitroprusside
(C)	Thiourea	(iii)	Blue color with hot and acidic solution of $FeSO_4$

(A) (A) - (ii); (B) - (i); (C) - (iii)

(B) (A) - (iii); (B) - (ii); (C) - (i)

(C) (A) - (ii); (B) - (iii); (C) - (i)

(D) (A) - (iii); (B) - (i); (C) - (ii)

Q80. JEE Main 2015 (Online) 10th April Morning Slot

Open solution

1.4 g of an organic compound was digested according to Kjeldahl's method and the ammonia evolved was absorbed in 60 mL of M/10 H_2SO_4 solution. The excess sulphuric acid required 20 mL of M/10 NaOH solution for neutralization. The percentage of nitrogen in the compound is :

(A) 3 **(B)** 5 **(C)** 10 **(D)** 24

Q81. JEE Main 2013 (Online) 22th April Morning Slot

Open solution

For which of the following compounds Kjeldahl method can be used to determine the percentage of Nitrogen?

(A) Nitrobenzene **(B)** Pyridine **(C)** Alanine **(D)** Diazomethane

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Q82. JEE Main 2013 (Online) 9th April Morning Slot

Which of the following compounds is not expected to show Lassaigne's test for nitrogen?

(A) Propanenitrile (B) Hydroxylamine hydrochloride

(C) Nitromethane (D) Ethanamine

Open solution

Q83. AIEEE 2012 (Online) 19th May

Beilstein test is used for the estimation of which one of the following elements?

(A) S (B) Cl (C) C and H (D) N

Open solution

Q84. AIEEE 2012 (Online) 7th May

Copper wire test for halogens is known as

(A) Dumas' Test (B) Beilstein's Test (C) Liebig's Test (D) Lassaigne's Test

Open solution

Q85. AIEEE 2011

Which of the following reagents may be used to distinguish between phenol and benzoic acid ?

(A) Tollen's reagent (B) Molisch reagent (C) Neutral $FeCl_3$ (D) Aqueous NaOH

Open solution

Q86. AIEEE 2010

29.5 mg of an organic compound containing nitrogen was digested according to Kjeldahl's method and the evolved ammonia was absorbed in 20 mL of 0.1 M HCl solution. The excess of the acid required 15 mL of 0.1 M NaOH solution for complete neutralization. The percentage of nitrogen in the compound is

(A) 59.0 (B) 47.4 (C) 23.7 (D) 29.5

Open solution

Q87. AIEEE 2004

The ammonia evolved from the treatment of 0.30 g of an organic compound for the estimation of nitrogen was passed in 100 mL of 0.1 M sulphuric acid. The excess of acid required 20 mL of 0.5 M sodium hydroxide solution for complete neutralization. The organic compound is

(A) urea (B) benzamide (C) acetamide (D) thiourea

Open solution

Q88. AIEEE 2004

The compound formed in the positive test for nitrogen with the Lassaigne solution of an organic compound is

(A) $Fe_4[Fe(CN)_6]_3$ (B) $Na_4[Fe(CN)_5NOS]$ (C) $Fe(CN)_3$ (D) $Na_3[Fe(CN)_6]$

Open solution

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Numerical 41 questions

Q89. JEE Main 2026 (Online) 5th April Morning Shift

In an estimation of sulphur by Carius method 0.2 g of the substance gave 0.6 g of $BaSO_4$. The percentage of sulphur in the substance is ____%.

(Given molar mass in $gmol^{-1}$ S : 32, $BaSO_4$: 231)

Open solution

Q90. JEE Main 2026 (Online) 4th April Evening Shift

In sulphur estimation, 2.0×10^{-3} mol of an organic compound (X) (molar mass 76 g mol^{-1}) gave 0.4813 g of barium sulphate (molar mass 233 g mol^{-1}). The percentage of sulphur in the compound (X) is ____ $\times 10^{-1}$% (Nearest integer)

Open solution

Q91. JEE Main 2026 (Online) 4th April Morning Shift

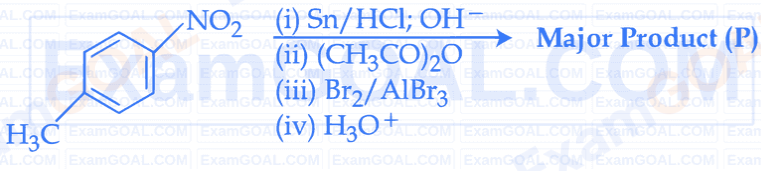
2.0 g of a bromo hydrocarbon (X) was subjected to Carius analysis, gave 3.36 g of AgBr . The percentage of carbon in the compound (X) is 26.7%. Total number of carbon atoms in the empirical formula for compound (X) is ____ .

(Given molar mass in $gmol^{-1}$ H : 1, C : 12, Br : 80, Ag : 108)

Open solution

Q92. JEE Main 2026 (Online) 2nd April Evening Shift

Consider the following reactions sequence



When the product (P) is subjected to Carius analysis using $AgNO_3$, 1.0 g of the product (P) will produce _____ g of the precipitate of AgBr. (Nearest Integer)

(Given : molar mass in g mol^{-1} C : 12, H : 1, O : 16, N : 14, Br : 80, Ag : 108)

Open solution

Q93. JEE Main 2026 (Online) 24th January Morning Shift

In Dumas method for estimation of nitrogen, 0.50 g of an organic compound gave 70 mL of nitrogen collected at 300 K and 715 mm pressure. The percentage of nitrogen in the organic compound is ____ %.

(Aqueous tension at 300 K is 15 mm).

Open solution

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Q94. JEE Main 2026 (Online) 22nd January Morning Shift

Sodium fusion extract of an organic compound (Y) with $CHCl_3$ and chlorine water gives violet color to the $CHCl_3$ layer. 0.15 g of (Y) gave 0.12 g of the silver halide precipitate in Carius method. Percentage of halogen in the compound (Y) is

____ . (Nearest integer)

(Given : molar mass $gmol^{-1} C : 12, H : 1, Cl : 35.5, Br : 80, I : 127$)

Open solution

Q95. JEE Main 2025 (Online) 7th April Evening Shift

In Dumas' method 292 mg of an organic compound released 50 mL of nitrogen gas (N_2) at 300 K temperature and 715 mm Hg pressure. The percentage composition of ' N ' in the organic compound is _____ % (Nearest integer) (Aqueous tension at $300\ K = 15\ mmHg$)

Open solution

Q96. JEE Main 2025 (Online) 4th April Morning Shift

In Dumas' method for estimation of nitrogen 1 g of an organic compound gave 150 mL of nitrogen collected at 300 K temperature and 900 mm Hg pressure. The percentage composition of nitrogen in the compound is _____ % (nearest integer)

(Aqueous tension at $300\ K = 15\ mm\ Hg$)

Open solution

Q97. JEE Main 2025 (Online) 3rd April Morning Shift

During estimation of nitrogen by Dumas' method of compound $X(0.42\ g)$



_____mL of N_2 gas will be liberated at STP. (nearest integer)

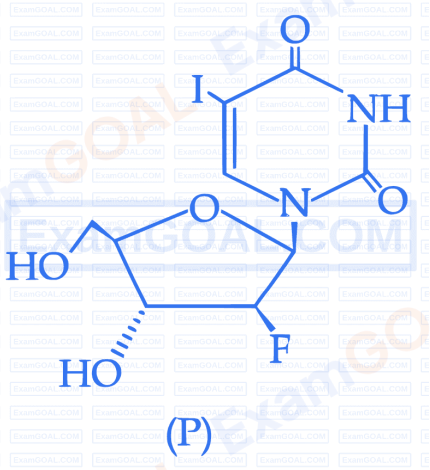
(Given molar mass in $g\ mol^{-1} : C : 12, H : 1, N : 14$)

Open solution

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Q98. JEE Main 2025 (Online) 2nd April Morning Shift

0.1 mol of the following given antiviral compound (P) will weigh _____ $\times 10^{-1}$ g (nearest integer).



(P)

(Given : molar mass in gmol^{-1} $\mathrm{H}: 1, \mathrm{C}: 12,\ \mathrm{N}: 14, \mathrm{O}: 16,\ \mathrm{F}: 19, \mathrm{I}: 127$)

Open solution

Q99. JEE Main 2025 (Online) 29th January Evening Shift

In the sulphur estimation, 0.20 g of a pure organic compound gave 0.40 g of barium sulphate. The percentage of sulphur in the compound is _____ $\times 10^{-1}\%$.

(Molar mass : $\mathrm{O} = 16,\ \mathrm{S} = 32, \mathrm{Ba} = 137$ in $\mathrm{g\ mol}^{-1}$)

Open solution

Q100. JEE Main 2025 (Online) 24th January Evening Shift

In Carius method of estimation of halogen, 0.25 g of an organic compound gave 0.15 g of silver bromide (AgBr). The percentage of Bromine in the organic compound is _____ $\times 10^{-1}\%$ (Nearest integer).

(Given : Molar mass of Ag is 108 and Br is $80\ \mathrm{g\ mol}^{-1}$)

Open solution

Q101. JEE Main 2025 (Online) 23rd January Morning Shift

During " S " estimation, 160 mg of an organic compound gives 466 mg of barium sulphate. The percentage of Sulphur in the given compound is _____ %.

(Given molar mass in $\mathrm{g\ mol}^{-1}$ of $\mathrm{Ba}: 137,\ \mathrm{S}: 32, \mathrm{O}: 16$)

Open solution

Q102. JEE Main 2024 (Online) 1st February Evening Shift

Following Kjeldahl's method, 1 g of organic compound released ammonia, that neutralised 10 mL of $2\mathrm{M}\ \mathrm{H_2SO_4}$. The percentage of nitrogen in the compound is _____ %.

Open solution

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Q103. JEE Main 2023 (Online) 13th April Evening Shift

$0.400\ \text{g}$ of an organic compound (X) gave $0.376\ \text{g}$ of AgBr in Carius method for estimation of bromine. $\%$ of bromine in the compound (X) is _____.

(Given: Molar mass $\text{AgBr} = 188\ \text{g mol}^{-1}$

$\text{Br} = 80\ \text{g mol}^{-1}$)

Open solution

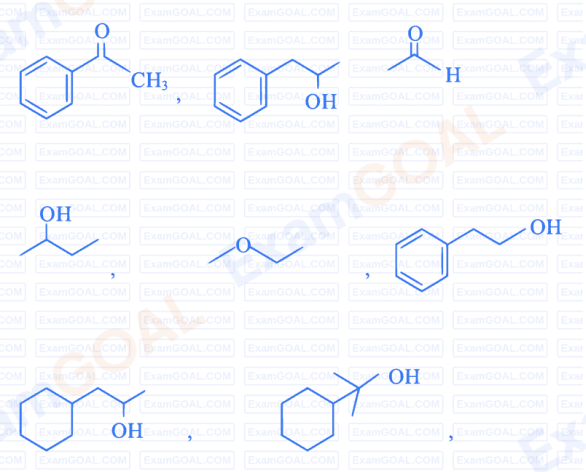
Q104. JEE Main 2023 (Online) 13th April Morning Shift

KMnO_4 is titrated with ferrous ammonium sulphate hexahydrate in presence of dilute H_2SO_4. Number of water molecules produced for 2 molecules of KMnO_4 is _____.

Open solution

Q105. JEE Main 2023 (Online) 25th January Evening Shift

Number of compounds giving (i) red colouration with ceric ammonium nitrate and also (ii) positive iodoform test from the following is _____



Open solution

Q106. JEE Main 2023 (Online) 25th January Morning Shift

In sulphur estimation, 0.471 g of an organic compound gave 1.4439 g of barium sulphate. The percentage of sulphur in the compound is _____ (Nearest Integer)

(Given : Atomic mass Ba: 137 u, S: 32 u, O: 16 u)

Open solution

Q107. JEE Main 2022 (Online) 29th July Morning Shift

In bromination of Propyne, with Bromine, 1, 1, 2, 2-tetrabromopropane is obtained in 27% yield. The amount of 1, 1, 2, 2-tetrabromopropane obtained from 1 g of Bromine in this reaction is _____ $\times 10^{-1}$ g. (Nearest integer)

(Molar Mass : Bromine = 80 g/mol)

Open solution

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Q108. JEE Main 2022 (Online) 28th July Evening Shift

A sample of $0.125\ \mathrm{g}$ of an organic compound when analyzed by Duma's method yields $22.78\ \mathrm{mL}$ of nitrogen gas collected over KOH solution at $280\ \mathrm{K}$ and $759\ \mathrm{mm\ Hg}$. The percentage of nitrogen in the given organic compound is _____. (Nearest integer)

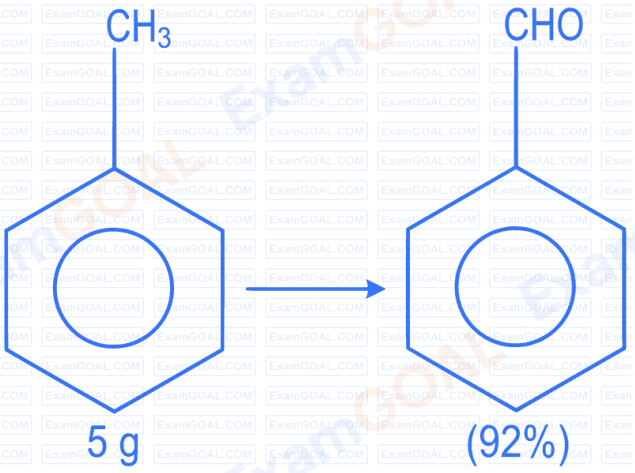
Given :

(a) The vapour pressure of water of $280\ \mathrm{K}$ is $14.2\ \mathrm{mm\ Hg}$.

(b) $\mathrm{R} = 0.082\ \mathrm{L\ atm\ K^{-1}\ mol^{-1}}$

Open solution

Q109. JEE Main 2022 (Online) 27th July Evening Shift



In the above reaction, $5\ \mathrm{g}$ of toluene is converted into benzaldehyde with 92% yield. The amount of benzaldehyde produced is _____ $\times 10^{-2}$ g. (Nearest integer)

Open solution

Q110. JEE Main 2022 (Online) 25th July Morning Shift

While estimating the nitrogen present in an organic compound by Kjeldahl's method, the ammonia evolved from $0.25\ \mathrm{g}$ of the compound neutralized $2.5\ \mathrm{mL}$ of $2\ \mathrm{M}\ \mathrm{H_2SO_4}$. The percentage of nitrogen present in organic compound is _____.

Open solution

Q111. JEE Main 2022 (Online) 30th June Morning Shift

An organic compound with 51.6% sulfur is heated in a Carius tube. The amount of this compound which will form 0.752 g of barium sulphate is _____ $\times\ 10^{-1}$ g.

(Given molar mass of barium sulphate 233 g $\mathrm{mol^{-1}}$) (Nearest integer).

Open solution

Q112. JEE Main 2022 (Online) 29th June Morning Shift

Kjeldahl's method was used for the estimation of nitrogen in an organic compound. The ammonia evolved from 0.55 g of the compound neutralised 12.5 mL of 1 M $\mathrm{H_2SO_4}$ solution. The percentage of nitrogen in the compound is _____. (Nearest integer)

Open solution

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Q113. JEE Main 2022 (Online) 28th June Morning Shift

A 2.0 g sample containing MnO_2 is treated with HCl liberating Cl_2. The Cl_2 gas is passed into a solution of KI and 60.0 mL of 0.1 M $Na_2S_2O_3$ is required to titrate the liberated iodine. The percentage of MnO_2 in the sample is _____. (Nearest integer)

[Atomic masses (in u) Mn = 55; Cl = 35.5; O = 16, I = 127, Na = 23, K = 39, S = 32]

Open solution

Q114. JEE Main 2022 (Online) 28th June Morning Shift

In the estimation of bromine, 0.5 g of an organic compound gave 0.40 g of silver bromide. The percentage of bromine in the given compound is _____ % (nearest integer)

(Relative atomic masses of Ag and Br are 108u and 80u, respectively).

Open solution

Q115. JEE Main 2022 (Online) 27th June Evening Shift

0.25 g of an organic compound containing chlorine gave 0.40 g of silver chloride in Carius estimation. The percentage of chlorine present in the compound is _____. [in nearest integer]

(Given : Molar mass of Ag is 108 g mol^{-1} and that of Cl is 35.5 g mol^{-1})

Open solution

Q116. JEE Main 2022 (Online) 24th June Evening Shift

0.2 g of an organic compound was subjected to estimation of nitrogen by Dumas method in which volume of N_2 evolved (at STP) was found to be 22.400 mL. The percentage of nitrogen in the compound is _____. [nearest integer]

(Given : Molar mass of N_2 is 28 g mol^{-1}. Molar volume of N_2 at STP : 22.4 L)

Open solution

Q117. JEE Main 2022 (Online) 24th June Morning Shift

A 0.166 g sample of an organic compound was digested with conc. H_2SO_4 and then distilled with NaOH. The ammonia gas evolved was passed through 50.0 mL of 0.5 N H_2SO_4. The used acid required 30.0 mL of 0.25 N NaOH for complete neutralization. The mass percentage of nitrogen in the organic compound is _____.

Open solution

Q118. JEE Main 2021 (Online) 31st August Evening Shift

The transformation occurring in Duma's method is given below :

$$C_2H_7N + \left(2x + \frac{y}{2}\right)CuO \rightarrow xCO_2 + \frac{y}{2}H_2O + \frac{z}{2}N_2 + \left(2x + \frac{y}{2}\right)Cu$$

The value of y is _____. (Integer answer)

Open solution

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Q119. JEE Main 2021 (Online) 27th August Morning Shift

In Carius method for estimation of halogens, 0.2 g of an organic compound gave 0.188 g of AgBr. The percentage of bromine in the compound is _____. (Nearest integer)

[Atomic mass : Ag = 108, Br = 80]

Open solution

Q120. JEE Main 2021 (Online) 27th August Morning Shift

The number of moles of CuO, that will be utilized in Dumas method for estimation of nitrogen in a sample of 57.5 g of N, N-dimethylaminopentane is _____ $\times 10^{-2}$. (Nearest integer)

Open solution

Q121. JEE Main 2021 (Online) 27th August Morning Shift

When 10 mL of an aqueous solution of $KMnO_4$ was titrated in acidic medium, equal volume of 0.1 M of an aqueous solution of ferrous sulphate was required for complete discharge of colour. The strength of $KMnO_4$ in grams per litre is _____ $\times 10^{-2}$. (Nearest integer)

[Atomic mass of K = 39, Mn = 55, O = 16]

Open solution

Q122. JEE Main 2021 (Online) 26th August Evening Shift

In the sulphur estimation, 0.471 g of an organic compound gave 1.44 g of barium sulphate. The percentage of sulphur in the compound is _____%. (Nearest integer)

(Atomic Mass of Ba = 137 u)

Open solution

Q123. JEE Main 2021 (Online) 27th July Morning Shift

An organic compound is subjected to chlorination to get compound A using 5.0 g of chlorine. When 0.5 g of compound A is reacted with $AgNO_3$ [Carius Method], the percentage of chlorine in compound A is _____ when it forms 0.3849 g of AgCl. (Round off to the Nearest Integer)

(Atomic masses of Ag and Cl are 107.87 and 35.5 respectively)

Open solution

Q124. JEE Main 2021 (Online) 25th July Evening Shift

0.8 g of an organic compound was analyzed by Kjeldahl's method for the estimation of nitrogen. If the percentage of nitrogen in the compound was found to be 42%, then _____ mL of 1 M H_2SO_4 would have been neutralized by the ammonia evolved during the analysis.

Open solution

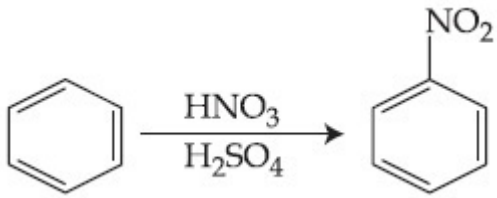
Q125. JEE Main 2021 (Online) 22th July Evening Shift

Methylation of 10 g of benzene gave 9.2 g of toluene. Calculate the percentage yield of toluene _____. (Nearest integer)

Open solution

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Q126. JEE Main 2021 (Online) 17th March Morning Shift



Open solution

In the above reaction, 3.9 g of benzene on nitration gives 4.92 g of nitrobenzene. The percentage yield of nitrobenzene in the above reaction is _____%. (Round off to the Nearest Integer).

(Given atomic mass : C : 12.0 u, H : 1.0 u, O : 16.0 u, N : 14.0 u)

Q127. JEE Main 2021 (Online) 16th March Evening Shift

Open solution

In Duma's method of estimation of nitrogen, 0.1840 g of an organic compound gave 30 mL of nitrogen collected at 287 K and 758 mm of Hg pressure. The percentage composition of nitrogen in the compound is _____. (Round off to the Nearest Integer). [Given : Aqueous tension at 287 K = 14 mm of Hg]

Q128. JEE Main 2021 (Online) 25th February Evening Shift

Open solution

Consider titration of NaOH solution versus 1.25 M oxalic acid solution. At the end point following burette readings were obtained.

(i) 4.5 mL

(ii) 4.5 mL

(iii) 4.4 mL

(iv) 4.4 mL

(v) 4.4 mL

If the volume of oxalic acid taken was 10.0 mL then the molarity of the NaOH solution is _____ M. (Rounded off to the nearest integer)

Q129. JEE Main 2020 (Online) 6th September Morning Slot

Open solution

In an estimation of bromine by Carius method, 1.6 g of an organic compound gave 1.88 g of AgBr. The mass percentage of bromine in the compound is _____.
(Atomic mass, Ag = 108, Br = 80 g mol^{-1})

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Out of syllabus 1 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q130. JEE Main 2021 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

Open solution

Which one of the following sets of elements can be detected using sodium fusion extract?

(A) Sulfur, Nitrogen, Phosphorus, Halogens

(B) Phosphorus, Oxygen, Nitrogen, Halogens

(C) Nitrogen, Phosphorus, Carbon, Sulfur

(D) Halogens, Nitrogen, Oxygen, Sulfur

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Answer Key

MCQ (Single Correct Answer)

1. **A**	2. **D**	3. **B**	4. **A**	5. **A**
6. **D**	7. **D**	8. **B**	9. **C**	10. **B**
11. **A**	12. **D**	13. **C**	14. **D**	15. **D**
16. **A**	17. **A**	18. **A**	19. **A**	20. **B**
21. **A**	22. **A**	23. **D**	24. **A**	25. **A**
26. **D**	27. **A**	28. **C**	29. **A**	30. **A**
31. **B**	32. **D**	33. **C**	34. **C**	35. **A**
36. **C**	37. **D**	38. **B**	39. **A**	40. **B**
41. **A**	42. **A**	43. **D**	44. **D**	45. **B**
46. **D**	47. **C**	48. **A**	49. **A**	50. **B**
51. **D**	52. **A**	53. **B**	54. **B**	55. **A**
56. **D**	57. **B**	58. **C**	59. **D**	60. **C**
61. **A**	62. **A**	63. **A**	64. **B**	65. **C**
66. **A**	67. **B**	68. **A**	69. **D**	70. **D**
71. **B**	72. **B**	73. **C**	74. **C**	75. **B**
76. **C**	77. **B**	78. **C**	79. **B**	80. **C**
81. **C**	82. **B**	83. **B**	84. **B**	85. **C**
86. **C**	87. **A**	88. **A**		

Numerical

89. **41 or 42**	90. **435**	91. **5**	92. **1**	93. **15**
94. **43**	95. **18**	96. **20**	97. **109**	98. **372**
99. **275**	100. **255**	101. **40**	102. **56**	103. **40**
104. **68**	105. **3**	106. **42**	107. **3**	108. **22**
109. **530**	110. **56**	111. **2**	112. **64**	113. **13**
114. **34**	115. **40**	116. **14**	117. **63**	118. **7**
119. **40**	120. **1125**	121. **316**	122. **42**	123. **19**
124. **12**	125. **78**	126. **80**	127. **19**	128. **6**
129. **50**				

Out of syllabus

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130. **A**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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